

RMSprop#

<https://pytorch.org/docs/stable/generated/torch.optim.RMSprop.html#torch.optim.RMSprop>

Description:

Implements RMSprop algorithm. For further details regarding the algorithm we refer to lecture notes by G. Hinton. and centered version Generating Sequences With Recurrent Neural Networks. The implementation here takes the square root of the gradient average before adding epsilon (note that TensorFlow interchanges these two operations). The effective learning rate is thus $\frac{\gamma}{(\bar{v} + \epsilon)} = \frac{\gamma}{\sqrt{\bar{v}} + \epsilon}$ where γ is the scheduled learning rate and $\bar{v}$ is the weighted moving average of the squared gradient. `params` (iterable) – iterable of parameters or `named_parameters` to optimize or iterable of dicts defining parameter groups. When using `named_parameters`, all parameters in all groups should be named `lr` (float, Tensor, optional) – learning rate (default: 1e-2) `alpha` (float, optional) – smoothing constant (default: 0.99)

Function Signature

```
class torch.optim.RMSprop(params, lr=0.01, alpha=0.99,  
eps=1e-08, weight_decay=0, momentum=0, centered=False,  
capturable=False, foreach=None, maximize=False,  
differentiable=False)[source]#
```

Parameters

```
add_param_group(param_group)[source]#
```

Parameters

```
load_state_dict(state_dict)[source]#
```

Returns:

dict[str, Any]

Code Examples

```

>>> model = torch.nn.Linear(10, 10)
>>> optim = torch.optim.SGD(model.parameters(), lr=3e-4)
>>> scheduler1 = torch.optim.lr_scheduler.LinearLR(
...     optim,
...     start_factor=0.1,
...     end_factor=1,
...     total_iters=20,
... )
>>> scheduler2 = torch.optim.lr_scheduler.CosineAnnealingLR(
...     optim,
...     T_max=80,
...     eta_min=3e-5,
... )
>>> lr = torch.optim.lr_scheduler.SequentialLR(
...     optim,
...     schedulers=[scheduler1, scheduler2],
...     milestones=[20],
... )
>>> lr.load_state_dict(torch.load("./save_seq.pt"))
>>> # now load the optimizer checkpoint after loading the
LRScheduler
>>> optim.load_state_dict(torch.load("./save_optim.pt"))

```

```
hook(optimizer) -> None
```

```
hook(optimizer, state_dict) -> state_dict or None
```

```
hook(optimizer, state_dict) -> state_dict or None
```

```
hook(optimizer) -> None
```

```
hook(optimizer, args, kwargs) -> None
```

```
hook(optimizer, args, kwargs) -> None or modified args and  
kwargs
```

```

{
  'state': {
    0: {'momentum_buffer': tensor(...), ...},
    1: {'momentum_buffer': tensor(...), ...},
    2: {'momentum_buffer': tensor(...), ...},
    3: {'momentum_buffer': tensor(...), ...}
  },
  'param_groups': [
    {
      'lr': 0.01,
      'weight_decay': 0,
      ...
      'params': [0]
      'param_names' ['param0'] (optional)
    },
    {
      'lr': 0.001,
      'weight_decay': 0.5,
      ...
      'params': [1, 2, 3]
      'param_names': ['param1', 'layer.weight',
'layer.bias'] (optional)
    }
  ]
}

```

Notes & Warnings

WARNING

Make sure this method is called after initializing `torch.optim.lr_scheduler.LRScheduler`, as calling it beforehand will overwrite the loaded learning rates.

NOTE

The names of the parameters (if they exist under the “param_names” key of each param group in `state_dict()`) will not affect the loading process. To use the parameters’ names for custom cases (such as when the parameters in the loaded state dict differ from those initialized in the optimizer), a custom `register_load_state_dict_pre_hook` should be implemented to adapt the loaded dict accordingly. If `param_names` exist in loaded state dict `param_groups` they will be saved and override the current names, if present, in the optimizer state. If they do not exist in loaded state dict, the optimizer `param_names` will remain unchanged.

Detailed Documentation

Documentation

Implements RMSprop algorithm.

For further details regarding the algorithm we refer to lecture notes by G. Hinton. and centered version Generating Sequences With Recurrent Neural Networks. The implementation here takes the square root of the gradient average before adding

epsilon (note that TensorFlow interchanges these two operations). The effective learning rate is thus $\gamma/(v+\epsilon)\gamma/(\sqrt{v} + \epsilon)$ where γ is the scheduled learning rate and v is the weighted moving average of the squared gradient.

params (iterable) – iterable of parameters or named_parameters to optimize or iterable of dicts defining parameter groups. When using named_parameters, all parameters in all groups should be named

lr (float, Tensor, optional) – learning rate (default: 1e-2)

alpha (float, optional) – smoothing constant (default: 0.99)

eps (float, optional) – term added to the denominator to improve numerical stability (default: 1e-8)

weight_decay (float, optional) – weight decay (L2 penalty) (default: 0)

momentum (float, optional) – momentum factor (default: 0)

centered (bool, optional) – if True, compute the centered RMSProp, the gradient is normalized by an estimation of its variance

capturable (bool, optional) – whether this instance is safe to capture in a graph, whether for CUDA graphs or for torch.compile support. Tensors are only capturable when on supported accelerators. Passing True can impair ungraphed performance, so if you don't intend to graph capture this instance, leave it False (default: False)

foreach (bool, optional) – whether foreach implementation of optimizer is used. If unspecified by the user (so foreach is None), we will try to use foreach over the for-loop implementation on CUDA, since it is usually significantly more performant. Note that the foreach implementation uses $\sim \text{sizeof}(\text{params})$ more peak memory than the for-loop

version due to the intermediates being a tensorlist vs just one tensor. If memory is prohibitive, batch fewer parameters through the optimizer at a time or switch this flag to False (default: None)

maximize (bool, optional) – maximize the objective with respect to the params, instead of minimizing (default: False)

differentiable (bool, optional) – whether autograd should occur through the optimizer step in training. Otherwise, the step() function runs in a torch.no_grad() context. Setting to True can impair performance, so leave it False if you don't intend to run autograd through this instance (default: False)

Add a param group to the Optimizer's param_groups.

This can be useful when fine tuning a pre-trained network as frozen layers can be made trainable and added to the Optimizer as training progresses.

param_group (dict) – Specifies what Tensors should be optimized along with group specific optimization options.

Load the optimizer state.

state_dict (dict) – optimizer state. Should be an object returned from a call to state_dict().

Make sure this method is called after initializing torch.optim.lr_scheduler.LRScheduler, as calling it beforehand will overwrite the loaded learning rates.

The names of the parameters (if they exist under the "param_names" key of each param group in state_dict()) will not affect the loading process. To use the parameters' names for custom cases (such as when the parameters in the loaded state dict differ from those initialized in the optimizer), a custom register_load_state_dict_pre_hook

should be implemented to adapt the loaded dict accordingly. If `param_names` exist in loaded state dict `param_groups` they will be saved and override the current names, if present, in the optimizer state. If they do not exist in loaded state dict, the optimizer `param_names` will remain unchanged.

Register a `load_state_dict` post-hook which will be called after `load_state_dict()` is called. It should have the following signature:

The optimizer argument is the optimizer instance being used.

The hook will be called with argument `self` after calling `load_state_dict` on `self`. The registered hook can be used to perform post-processing after `load_state_dict` has loaded the `state_dict`.

`hook` (Callable) – The user defined hook to be registered.

`prepend` (bool) – If `True`, the provided post hook will be fired before all the already registered post-hooks on `load_state_dict`. Otherwise, the provided hook will be fired after all the already registered post-hooks. (default: `False`)

a handle that can be used to remove the added hook by calling `handle.remove()`

`torch.utils.hooks.RemoveableHandle`

Register a `load_state_dict` pre-hook which will be called before `load_state_dict()` is called. It should have the following signature:

The optimizer argument is the optimizer instance being used and the `state_dict` argument is a shallow copy of the `state_dict` the user passed in to `load_state_dict`. The hook may modify the `state_dict` inplace or optionally return a new one. If a `state_dict` is returned, it will be used to be loaded into the optimizer.

The hook will be called with argument `self` and `state_dict` before calling `load_state_dict` on `self`. The registered hook can be used to perform pre-processing before the `load_state_dict` call is made.

`hook (Callable)` – The user defined hook to be registered.

`prepend (bool)` – If `True`, the provided pre hook will be fired before all the already registered pre-hooks on `load_state_dict`. Otherwise, the provided hook will be fired after all the already registered pre-hooks. (default: `False`)

a handle that can be used to remove the added hook by calling `handle.remove()`

`torch.utils.hooks.RemoveableHandle`

Register a state dict post-hook which will be called after `state_dict()` is called.

It should have the following signature:

The hook will be called with arguments `self` and `state_dict` after generating a `state_dict` on `self`. The hook may modify the `state_dict` inplace or optionally return a new one. The registered hook can be used to perform post-processing on the `state_dict` before it is returned.

`hook (Callable)` – The user defined hook to be registered.

`prepend (bool)` – If `True`, the provided post hook will be fired before all the already registered post-hooks on `state_dict`. Otherwise, the provided hook will be fired after all the already registered post-hooks. (default: `False`)

a handle that can be used to remove the added hook by calling `handle.remove()`

`torch.utils.hooks.RemoveableHandle`

Register a state dict pre-hook which will be called before `state_dict()` is called.

It should have the following signature:

The optimizer argument is the optimizer instance being used. The hook will be called with argument `self` before calling `state_dict` on `self`. The registered hook can be used to perform pre-processing before the `state_dict` call is made.

hook (Callable) – The user defined hook to be registered.

prepend (bool) – If True, the provided pre hook will be fired before all the already registered pre-hooks on `state_dict`. Otherwise, the provided hook will be fired after all the already registered pre-hooks. (default: False)

a handle that can be used to remove the added hook by calling `handle.remove()`

`torch.utils.hooks.RemoveableHandle`

Register an optimizer step post hook which will be called after optimizer step.

It should have the following signature:

The optimizer argument is the optimizer instance being used.

hook (Callable) – The user defined hook to be registered.

a handle that can be used to remove the added hook by calling `handle.remove()`

`torch.utils.hooks.RemovableHandle`

Register an optimizer step pre hook which will be called before optimizer step.

It should have the following signature:

The optimizer argument is the optimizer instance being used. If args and kwargs are modified by the pre-hook, then the transformed values are returned as a tuple containing the new_args and new_kwargs.

hook (Callable) – The user defined hook to be registered.

a handle that can be used to remove the added hook by calling handle.remove()

torch.utils.hooks.RemovableHandle

Return the state of the optimizer as a dict.

It contains two entries:

differs between optimizer classes, but some common characteristics hold. For example, state is saved per parameter, and the parameter itself is NOT saved. state is a Dictionary mapping parameter ids to a Dict with state corresponding to each parameter.

parameter group is a Dict. Each parameter group contains metadata specific to the optimizer, such as learning rate and weight decay, as well as a List of parameter IDs of the parameters in the group. If a param group was initialized with named_parameters() the names content will also be saved in the state dict.

NOTE: The parameter IDs may look like indices but they are just IDs associating state with param_group. When loading from a state_dict, the optimizer will zip the param_group params (int IDs) and the optimizer param_groups (actual nn.Parameter s) in order to match state WITHOUT additional verification.

A returned state dict might look something like:

Perform a single optimization step.

closure (Callable, optional) – A closure that reevaluates the model and returns the loss.

Reset the gradients of all optimized `torch.Tensor`s.

`set_to_none` (bool, optional) –

Instead of setting to zero, set the grads to None. Default: True

This will in general have lower memory footprint, and can modestly improve performance. However, it changes certain behaviors. For example:

When the user tries to access a gradient and perform manual ops on it, a None attribute or a Tensor full of 0s will behave differently.

If the user requests `zero_grad(set_to_none=True)` followed by a backward pass, `.grads` are guaranteed to be None for params that did not receive a gradient.

`torch.optim` optimizers have a different behavior if the gradient is 0 or None (in one case it does the step with a gradient of 0 and in the other it skips the step altogether).